

RESEARCH ARTICLE

Delineating ecologically distinct groups for annual cycle management of a declining shorebird

Elly C. Knight^{1,2}  | Jay Carlisle³ | Andy J. Boyce¹ | David Bradley⁴  |
 Paula Cimprich⁵  | Stephanie Coates⁶ | Stephen J. Dinsmore⁷ | Cory J. Gregory⁸ |
 Joel G. Jorgensen⁹ | Jeffrey F. Kelly⁵ | David Newstead¹⁰ | Alina Olalla¹¹ |
 Larkin A. Powell¹² | Amy L. Scarpignato¹ | T. Lee Tibbitts¹³ | Nils Warnock¹⁴ |
 Walter Wehtje¹⁵ | Peter P. Marra¹⁶ | Autumn-Lynn Harrison¹

Correspondence

Elly C. Knight
 Email: ecknight@ualberta.ca

Funding information

U.S. Bureau of Land Management; BC Hydro; John and Adrienne Mars; American Prairie; California Department of Fish and Wildlife; Meg and Bert Raynes Wildlife Fund; Habitat Conservation Trust Foundation; Columbia Basin Trust; McDanel Land Foundation; Page Family Foundation; Wyoming Governor's Office Big Game License Coalition; ConocoPhillips; Environment and Climate Change Canada; Nature Conservancy; MPG Ranch; U.S. Fish and Wildlife Service; Wyoming Game and Fish Department

Handling Editor: Tien Ming Lee

Abstract

1. Patterns of migratory connectivity are increasingly used to understand and manage threats throughout the annual cycle of migratory species. Strong migratory connectivity refers to when individuals from different populations remain spatially separated across the annual cycle, which may expose populations to unique sets of threats and conditions that cause differential population trends. However, the populations or groups used for species' management are often defined a priori based on expert knowledge and/or management units, which may mask important population segregation and obscure differential population trends and their drivers.
2. We compared three approaches to defining management groups of a declining shorebird, the long-billed curlew (*Numenius americanus*), for annual cycle management: by expert-opinion, according to management flyways, and with unsupervised clustering of satellite tracking data that maximizes the strength of migratory connectivity.
3. Despite the curlews having a continuous breeding range and a pattern of parallel migration, all three approaches identified groups with different population trends, movement behaviours and habitat selection across the annual cycle, suggesting these are meaningful ecological groups. The expert and clustering approaches resulted in similar group structure, strong estimates of migratory connectivity (measured as $MC = 0.64$ across seasons), movement behaviour and habitat selection; however, the expert approach identified an additional divide between the easternmost grouping, which revealed strongly negative population trends in the group occupying the Chihuahuan desert during the stationary nonbreeding season. In contrast, the flyway delineation resulted in weaker estimates of migratory connectivity, marginal differences in population trends and less between-group differences in movement behaviour and habitat selection.

For affiliations refer to page 1162.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 Smithsonian Institution and The Author(s). *Journal of Applied Ecology* published by John Wiley & Sons Ltd on behalf of British Ecological Society. This article has been contributed to by U.S. Government employees and their work is in the public domain in the USA.

4. *Synthesis and applications.* Using measurements of migratory connectivity in concert with expert opinion can define ecologically distinct groups for wildlife management that differ in the environmental conditions they experience across seasons of the annual cycle, which is a key component for understanding and reversing declines of migratory species.

KEYWORDS

curlew, habitat selection, migratory connectivity, movement ecology, *Numenius*, population, wildlife management

1 | INTRODUCTION

The spatial structure of migratory populations is dynamic as individuals move among the various locations of their annual cycle. Understanding the resultant pattern, known as migratory connectivity, can provide important insights into the factors structuring underlying population dynamics (Marra et al., 2018; Webster et al., 2002). When migratory connectivity is strong, individuals from different populations in one stage of the annual cycle remain spatially separated during other stages and may experience unique sets of environmental conditions year-round (Cresswell, 2014). When migratory connectivity is weak and different populations mix during other stages, effects of environmental conditions in those stages are distributed across the species' range. In addition to direct population-level effects, the strength of migratory connectivity can also be related to a species' ability to adapt to a changing environment (Cresswell, 2014; Iwamura et al., 2013) and result in population differentiation and speciation (Justen et al., 2021; Møller & Szép, 2011; Ruegg & Smith, 2002).

In a management context, determining the pattern of migratory connectivity is an important step towards understanding differential population trends (Cresswell, 2014; Rushing et al., 2016; Webster et al., 2002). When populations use different migration routes, differential trends can potentially be attributed to the pattern of migratory connectivity via the exposure to different environmental conditions at some point during the annual cycle (Anderwald et al., 2021; Morrick et al., 2022). Despite this attribution, differential population trends are rarely quantitatively evaluated in a migratory connectivity framework per se (but see Hewson et al., 2016; Kramer et al., 2018; Rushing et al., 2016).

Many studies of population divides focus on spatially disjunct populations (e.g. Anderwald et al., 2021; Procházka et al., 2018), despite their existence in populations with continuous breeding ranges (but see Hobson et al., 2015; Finch et al., 2015). Examining migratory connectivity across a species' full range can identify biologically important, but otherwise hidden, divides, where individuals occupy the same relative longitude across their annual cycle (Alerstam & Hedenström, 1998). In contrast, a priori approaches to identifying groups using administrative flyways (Boere & Stroud, 2006) or natural populations (Rushing et al., 2015) may not allow for the identification of true population divides across the annual cycle. Such a priori

groups may result in an underestimation of the strength of migratory connectivity, as suggested by simulation (Vickers et al., 2021), or an overestimation when using larger regions such as flyways. Therefore, a priori approaches could potentially lead to obscured differential population trends and a misunderstanding of the drivers of those trends.

Identifying population divides to delineate ecologically distinct groups for management is therefore a critical first step for conservation because it enables the comparison of population trends and subsequent identification and reversal of limitations for declining populations (Rushing et al., 2016). Shorebirds (*Charadriiformes*), a rapidly declining group of waterbird species (Pearce-Higgins et al., 2017; Smith, Binley, et al., 2023; Smith, Smith, et al., 2023), could benefit from identification of such groupings for full-annual cycle management. Many North American shorebird species have continuous breeding ranges that span large portions of the continent. However, unlike avian taxa with weak migratory connectivity that are therefore unlikely to have population divides (Finch et al., 2017), many shorebird species have strongly parallel migration that results in strong migratory connectivity (e.g. Huysman et al., 2022). Furthermore, many shorebird species have high inter-annual site fidelity to discrete and often imperilled areas used for lengthy stopovers during migration, and so understanding how populations vary in stopover use is important (Warnock, 2010).

The long-billed curlew (*Numenius americanus*) is North America's largest shorebird species, and widely considered a species of conservation concern due primarily to extensive habitat loss and degradation throughout their range (Dugger & Dugger, 2020). They are part of the *Numeniini* tribe of shorebirds, a group of 13 species that is globally recognized as imperilled (Pearce-Higgins et al., 2017). Though monitoring data suggest a neutral overall trend (Sauer et al., 2017), community science (also known as 'citizen science') suggests populations are declining in many regions (Fink, Johnston, et al., 2023), and the species has been extirpated from the eastern portion of the historic breeding range (Dugger & Dugger, 2020).

Our goal was to delineate ecologically distinct groups of long-billed curlews for annual cycle conservation management (Marra et al., 2015). We collated existing satellite tracking datasets of long-billed curlews collected across their range by the authors and clustered the geographic positions of individual birds across the annual cycle. We compared three approaches to delineating groups: flyway units

for shorebird conservation (Roberts et al., 2023), expert knowledge and an unsupervised clustering approach (hereinafter 'flyway', 'expert' and 'clustering'). We used a clustering approach that maximized the strength of migratory connectivity to select the number of groups because if population trend is linked to differences in environmental conditions driven by migratory connectivity, then selecting groups by maximizing the strength of migratory connectivity should result in groups with varying population trends. Accordingly, we then estimated the population trend of each group for each approach using data from the North American Breeding Bird Survey. Finally, we evaluated whether those groups differed in movement behaviours (e.g. migration timing, home range size per season) and seasonal habitat selection strength for several environmental attributes to test for meaningful ecological differences between groups and to help guide next steps for research and management of this imperilled species.

2 | MATERIALS AND METHODS

2.1 | Datasets

We collated long-billed curlew tracking data collected via Argos satellite or GPS transmitters from 2007 to 2021 (Olalla-Kerstupp et al., 2015a, 2015b; Page et al., 2014; Coates, 2018; Movebank IDs 25608436, 46195209, 1377708730, 1418296656, 1953194738; Kays et al., 2022; Figure 1; Appendix S1). Individual adult birds were

captured using mist-nets placed over incubating birds, cannon nets, or leg-hold noose mats. Tag masses ranged from 9.5 to 22 g (<3.5% body mass) and were affixed with Teflon ribbon leg-loop harnesses (Page et al., 2014). Tags that collected Argos locations (CLS America, 2016) were typically programmed to cycle for 5 or 10 h on and 24 h off. GPS tag schedules depended on battery power levels and tag model but were typically at least 1 location per hour. All animal capture and tag deployment was conducted under Institutional Animal Care and Use Committee approvals from Montana Fish, Wildlife and Parks (FWP7-2009), Iowa State University (3-08-6520-Q), Boise State University (006-AC11-004, 006-AC14-008, 006-AC17-005, AC20-008) and the Smithsonian National Zoological Park (14-30, 18-06, 19-07); under federal bird banding or scientific collection permits from the United States Geological Survey (09316, 23285, 22929-A, 24243), Environment and Climate Change Canada (10904, 10904 A), and the México Secretaría de Medio Ambiente y Recursos Naturales (01989, 03850); under state or provincial permits from Montana Fish, Wildlife and Parks (2009-044, 2016-084, 2017-096, 2019-068, 2020-058-W), Oregon Department of Fish and Wildlife Scientific (040-07, 059-08), Nevada Department of Wildlife (29559), Nebraska Game and Parks Commission (993), British Columbia Ministry of Forest, Land and Natural Resource (CB17-271056, 100205517), Idaho (990121), Wyoming (33-943), New Mexico (3728); and under special use permits from the Rio Mora National Wildlife Refuge (2020_04R, 2021_06R), National Elk Refuge (CurlewsIBO2017), and Grand Teton National Park (GRTE-2015-SCI-0047, GRTE-2016-SCI-0019, GRTE-2017-SCI-044).

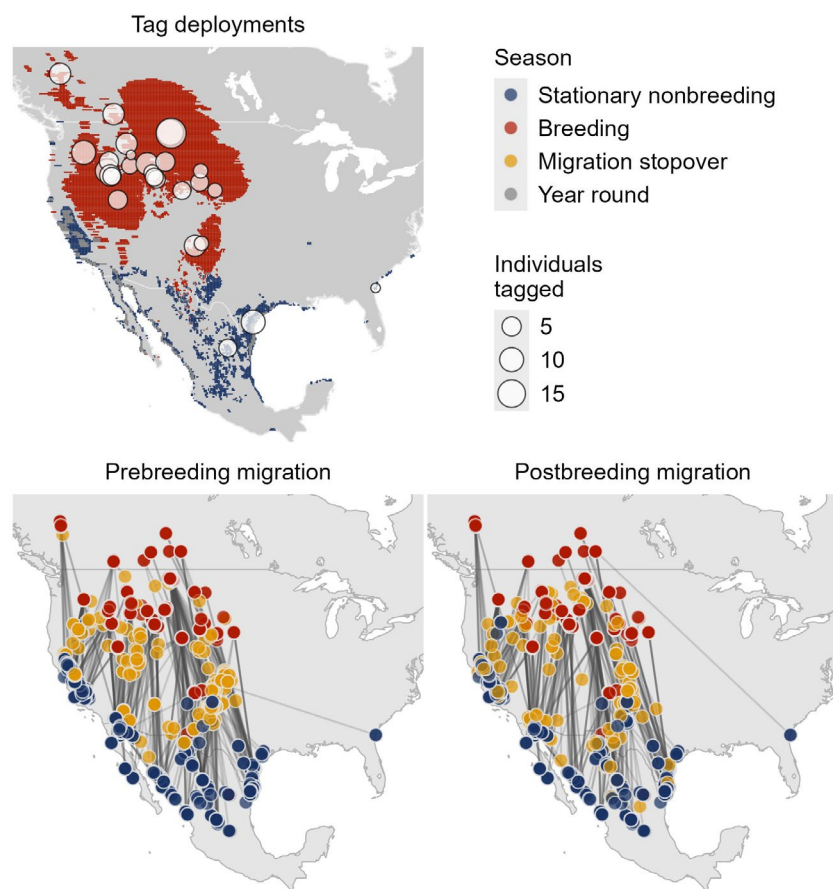


FIGURE 1 Location of sites where satellite tags were deployed on individual long-billed curlews during nine studies across North America, 2007–2021, and resultant segmented locations linked by individuals across seasons. Seasonal range data from eBird (Fink, Auer, et al., 2023).

2.2 | Movement data processing

We combined the raw data from all nine tracking datasets to create a single, standardized dataset (Appendices S1, S2 and S3). We used a random walk model in the *foieGras* package (Jonsen et al., 2020) to (1) estimate probable movement paths for individual birds, (2) standardize sampling rates across individuals and (3) account for telemetry error (typically ranges from 500 to 2500 m for Argos data; CLS America, 2016). We predicted locations at 24-h intervals and removed predictions for transmission gaps longer than 2 weeks to avoid inaccurate location predictions and misidentification of migration departure or arrival dates. We then used a multi-step classification process that included net-squared displacement (NSD) and a behavioural two-state hidden Markov model (HMM) in *move HMM* package (Michelot et al., 2016) to segment each individual's predicted daily locations into four seasons per year: breeding, postbreeding migration, nonbreeding stationary (hereinafter 'nonbreeding') and prebreeding migration. We used another two-state HMM to identify stopovers during migration that were at least 2 days long, and a third two-state HMM to separate multiple nonbreeding home ranges. We removed one individual that spent the nonbreeding season on the Atlantic coast from further analysis because, although there remains a very small population of long-billed curlews that winter in this region, the small sample size from that outlier had undue influence on our analysis.

2.3 | Group delineation approach

2.3.1 | Flyway groups

We adapted existing groupings that assign each US state and Canadian province to a flyway based on geography and waterfowl harvest management population models (Roberts et al., 2023; US Fish and Wildlife Service, 2019). We combined the Central and Mississippi flyways into a single Midcontinent flyway based on recent initiatives to guide shorebird management and conservation (e.g. Midcontinent Shorebird Conservation Initiative, Shorebird Science and Conservation Collective) and the identification of the importance of the Midcontinent flyway for other groups of birds (Knight et al., 2018; Lott & Smith, 2006). We then assigned each of the long-billed curlews in our dataset to one of the three resultant flyway groups ('Pacific', 'Midcontinent', 'Atlantic') based on their breeding location.

2.3.2 | Expert-defined groups

We used a priori knowledge of long-billed curlew ecology (Dugger & Dugger, 2020) to separate long-billed curlews into groups across their range (Figure 2). We separated our tracked individuals into four groups based on their ecology and geography during the stationary

nonbreeding season, particularly in the eastern portion of the wintering range where birds occupying coastal marshes along the gulf coast experience different conditions than birds in the high-elevation Chihuahuan desert.

2.3.3 | Clustering groups

We used the expert-defined groups to determine the potential number of groups for our quantitative delineation (i.e. up to one more than the number of expert-defined groups). We clustered breeding, migration stopover and nonbreeding spatial locations into two to five groupings and then used migratory connectivity estimates to select the number of groups that would maximize migratory connectivity. We selected this season-specific approach after preliminary exploration of clustering continuous movement tracks with dynamic time warping (Giorgino, 2009), which did not yield contiguous clusters of individuals. First, we calculated unique seasonal sites for each individual in each year as the mean of the daily locations for each breeding home range, nonbreeding home range, and migration stopover. We then selected the site for each individual in each season that was occupied for the longest duration (i.e. number of days). Preliminary analyses showed this provided similar results to random selection and bootstrapping across the varying numbers of years of data and numbers of stopover and nonbreeding sites across individuals. We used incomplete K-means cluster analysis in the *ClustImpute* package (Pfaffel, 2020) to estimate groups of individuals that were spatially clustered across the annual cycle. We selected the number of groups that maximized mean migratory connectivity strength (MC; see below) across all six pairwise comparisons.

2.3.4 | Migratory connectivity estimation

Migratory connectivity strength (MC) is quantified as the correlation of population configuration between two seasons on a scale of 0 to 1 (weak to strong respectively). We estimated MC using the *MigConnectivity* package (Cohen et al., 2017) between the breeding grounds and each of the other three seasons of the annual cycle for each approach. To avoid a breeding-ground-centric perspective, we also estimated migratory connectivity between the stationary nonbreeding grounds and the other three seasons of the annual cycle. We chose to quantify migratory connectivity using MC over other metrics (e.g. Mantel test; Ambrosini et al., 2009), because MC incorporates cluster membership, which we were specifically interested in. MC also incorporates transition probabilities between clusters between seasons, which subsequently allows for the inclusion of uncertainty due to sample size, detection heterogeneity and location error by resampling the transition probabilities from a normal distribution 1000 times. To estimate MC, we defined origins as a 500 km buffer around the centroid of each group of locations per group. We defined targets by dividing the range of the target season into a 500 km grid and we assigned target values to each individual based

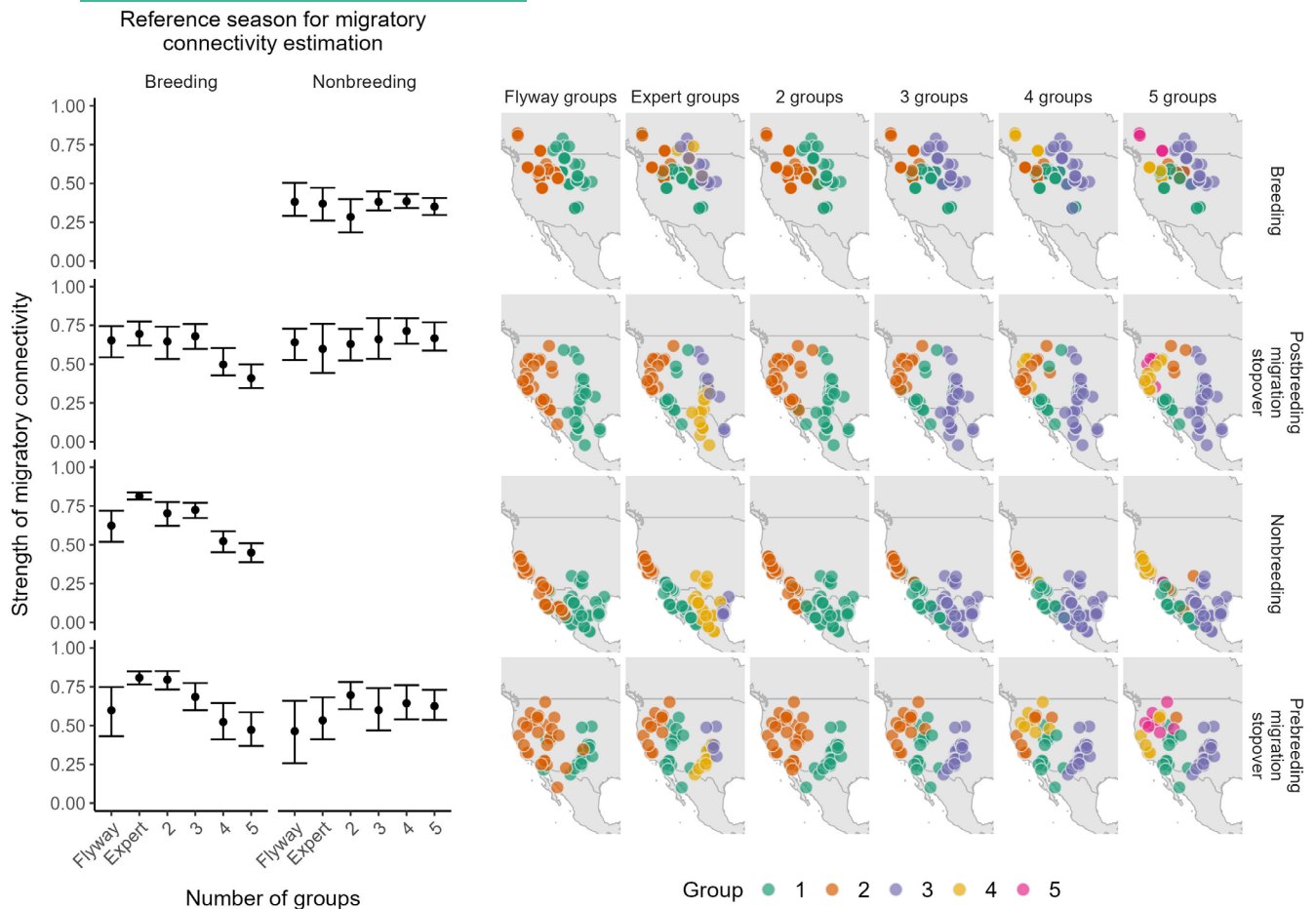


FIGURE 2 Spatial migratory connectivity (left) and group assignment (right) of individual long-billed curlews across the annual cycle according to different group delineation approaches (flyway, expert, clustering from 2 to 5 groups). Error bars represent the standard error of the estimates.

on grid squares (Hallworth et al., 2021). To incorporate information on relative abundance within each group that is used to modify transition probabilities, we extracted the point level value of estimated mean relative abundance from the eBird abundance map (Sullivan et al., 2009) for the appropriate origin season (breeding, nonbreeding) using the ebirdst package (Strimas-Mackey et al., 2023) and calculated the mean across all points (i.e. individuals).

2.4 | Population trend

We used data from 1970 to 2019 of the North American Breeding Bird Survey (BBS) to estimate population trend for each selected group in each approach (Ziolkowski et al., 2023). We included all BBS routes with at least one long-billed curlew detection that were within 500km from the breeding location of a tracked individual (the 95th quantile of those distances). We assigned each BBS route to a group using k -nearest neighbour (KNN) models ($k=10$) trained with the group assignments and breeding coordinates of the tracked individuals. We trained one KNN and predicted the group of each BBS route.

We estimated population trend from the BBS data for each group using the bbsBayes package (Smith & Edwards, 2020). We used a generalized additive model with random intercept year effects (GAMYE), which allows for nonlinear population trajectories while also accounting for interannual variation. We calculated population trend from the GAMYE trajectory for the period of 2007–2019 to match the time period of our tracking dataset.

2.5 | Movement behaviour

We quantified seven movement behaviours that might represent ecological differences and provide insight into population trends across groups: migration departure and arrival timing, daily displacement distance (km/day from great circle distance), number of migration stopovers, migration stopover duration (total days spent at stopovers within a season), number of wintering home ranges, and core stopover or home range size (50% isopleth of use distribution from kernel density estimation). We used the adehabitatHR package (Calenge, 2006) with the ad hoc method of defining the smoothing parameter and a bivariate normal kernel to estimate core stopover or home range size. We

calculated each behaviour for all relevant seasons for each combination of individual and year that we had tracking data for, with separate values within a season for individuals with multiple migration stopovers or nonbreeding grounds (Appendices S3 and S4).

For each group delineation approach, we tested for differences in movement behaviours among long-billed curlew groups using linear mixed effects models in the package lme4 (Bates et al., 2015). We built separate global models for each movement behaviour with season, group and an interaction between the two as fixed effects, and individual ID as a random effect. We compared each global model to all simpler models with Akaike's information criterion (AIC; Akaike, 1987) and interpreted the fixed covariates of the most parsimonious model within Δ AIC of two as having an effect on the behavioural attribute for that model.

2.6 | Habitat selection

We tested for differences among groups in second order (i.e. home range; Johnson, 1980) habitat selection for each delineation approach. We defined used locations as the centroid of each individual's daily locations for each home range or stopover location in each year. We defined the availability domain (i.e. the area that the animal can choose from given its current location) as the 95% minimum convex polygon (MCP) of the used locations for each combination of group and season and randomly selected 10 available locations within those MCPs for each used location.

For each used or available location, we extracted three environmental attributes: wetland proportion, cropland proportion and grassland proportion (Cochran & Anderson, 1987; Fellows & Jones, 2009). We extracted proportions from the annual MODIS landcover type dataset (Friedl & Sulla-Menashe, 2022), with temporal matching to the year of tracking data using the rgee package (Aybar, 2024). We extracted each covariate as the mean within a buffer around each used or available point with the same radius as the mean stopover or home range area for that combination of season and group. We checked the three environmental attributes for excessive covariation in each season by quantifying the variance inflation factor (VIF) with the usdm package (Naimi et al., 2013) and found VIF values were less than 3.5 for all attributes in all seasons.

For each delineation approach, we tested for differences in habitat selection of long-billed curlew groups for each environmental attribute using logistic regression resource selection functions (Manly et al., 2002). We initially used a mixed effects model with individual as a random effect on intercept to account for any potential pseudoreplication due to repeat sampling of individuals across years (Appendix S1, Table S1.2); however, the random effect explained almost no variation in the model, so we instead used a simpler fixed-effects only model. For each season, we created a global model with used or available as the response and all four environmental attributes as predictors as well as an interaction between each attribute and group to test for differences in selection between

groups. We compared the global model for each season to all simpler models with AIC and selected the most parsimonious model within Δ AIC of two. All analyses were conducted in R version 4.2.1 (R Core Team, 2021).

3 | RESULTS

The compiled dataset comprised 98,720 daily locations from 130 individuals (270.5 bird years; Figure 1). The mean number of locations per individual was 759 ± 649 SD (2.08 ± 1.78 SD years) and the range was 60 to 4450 locations (0.16–12.20 years). Locations were summarized to 345 unique breeding and 412 nonbreeding home ranges. There were 166 unique postbreeding and 210 prebreeding migration stopover sites across individuals that made stops in each migration season (81, 75 individuals in the postbreeding and prebreeding season, respectively).

Long-billed curlews exhibited a parallel migration strategy, with individuals typically occupying migration stopover and nonbreeding home ranges of similar longitude to their breeding home range, with the exception of an individual that spent the breeding season in Saskatchewan and the nonbreeding season in Georgia (Figure 1). The tracked long-billed curlews showed complex migration ecology; there was variation across and even within individuals in whether migration stopovers were recorded, the duration of stopover, migration speed, the use of multiple nonbreeding home ranges and movements associated with pre-migratory staging.

3.1 | Group delineation

The flyway and expert approaches created visibly contiguous groups of approximately equal size, as did the clustering approach with up to three groups (Figure 2). The flyway approach created groups that partially overlapped during the nonbreeding and prebreeding migration seasons, while the expert approach created groups that were partially spatially overlapped during the breeding season. The spatial segregation of groups for the clustering approach appeared to decrease with increasing number of groups (i.e. K clusters).

3.2 | Migratory connectivity

The estimated strength of migratory connectivity (MC) was strong; mean MC across season comparisons, delineation approaches, and numbers of groups was 0.58 (0.49–0.68 SE). MC was greater than 0.60 for all six pairwise season comparisons, except for nonbreeding to breeding MC, which was moderate ($MC = 0.36 \pm 0.12$ SE; Figure 2).

Mean MC across seasons was strongest for the expert approach (0.64 ± 0.09 SE), partially because MC from breeding to nonbreeding locations was particularly strong (0.81 ± 0.03 SE; Figure 2). Mean MC for the flyway approach was 0.56 (± 0.12 SE),

with particularly weak estimates between the breeding season and both the nonbreeding and prebreeding migration seasons relative to the other approaches. MC for the clustering approach was affected by the number of groups, with MC generally decreasing with an increasing number of groups, although there was strong overlap in 95% CIs across the number of groups. The strongest MC for the clustering approach was for two or three groups (0.62 ± 0.09 SE). We selected three as the number of groups for the clustering approach because there was lower variance in MC estimates across seasons than with two groups and therefore better maximized MC across all seasons. The three clustered groups were similar to our expert groups (Figure 2); however, we did not find support for the easternmost divide between individuals in the eastern group that wintered in inland versus coastal areas. Migratory connectivity did not delineate between these groups, likely because the tracking data revealed there was movement between the two groups within seasons by individual birds.

We hereinafter refer to the groups for the flyway approach as per the flyway names of 'Pacific' and 'Midcontinent'; the groups for the expert approach as 'West', 'Intermountain', 'Chihuahuan' and 'Gulf'; and the groups for the clustering approach as 'West', 'Intermountain' and 'Plains' (Figure 3; Appendix S4; Knight et al., 2025).

3.3 | Population trend

Long-billed curlews were detected on 521 BBS routes between 1970 and 2019, of which 494 were included in our analysis based on proximity to the breeding home ranges of our tracked individuals (Figure 3). The mean number of curlews detected per route was $2.77 (\pm 6.70$ SD; range 1–117). All three delineation approaches showed differences in population trend between groups; however, the difference between the Pacific and Midcontinent groups in the flyway approach was marginal. The flyway approach suggested declines for the Midcontinent group and marginally positive population trends for the Pacific group; however, the expert and clustering approaches revealed that the far west portion of the Pacific group was also declining. The expert approach further revealed that the declines in the east were influenced by a strong negative trend for the Chihuahuan group, while the Gulf group population trend was positive.

3.4 | Movement behaviour

The patterns in movement behaviour between groups were similar for all three delineation approaches, with the exception of departure timing, for which there was no between group difference using the flyway

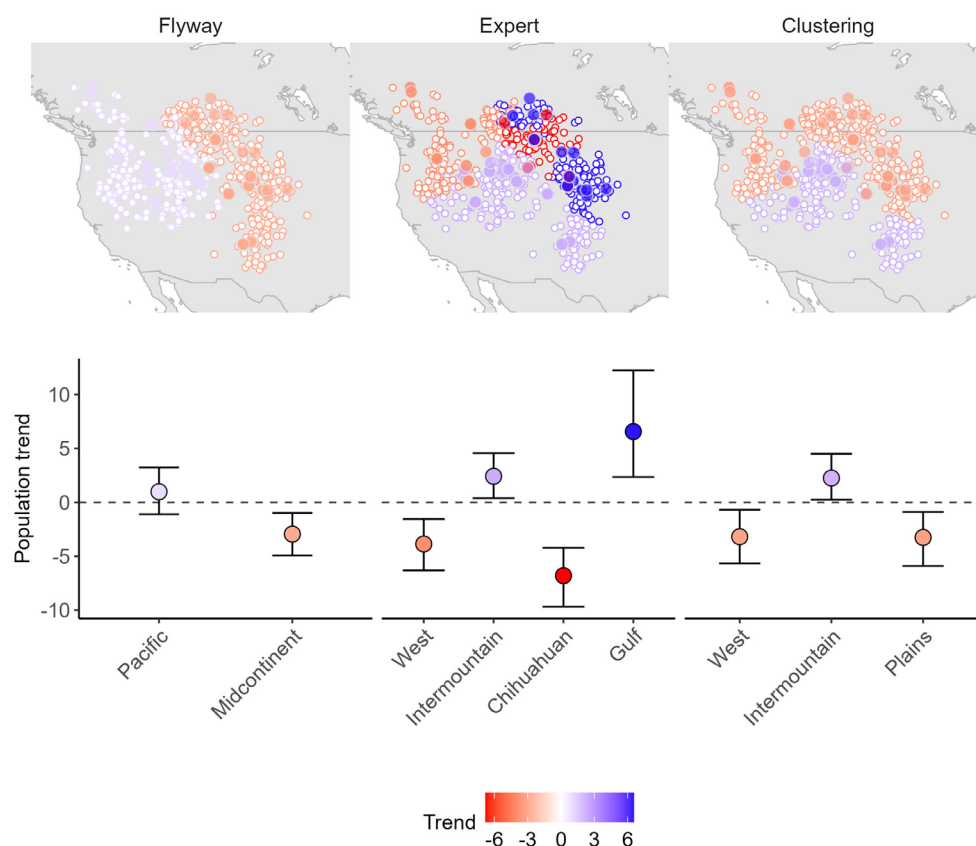


FIGURE 3 Top: Assigned group of individual long-billed curlew breeding home range locations (solid colour) used to train a *k*-nearest neighbour model for classification BBS routes with long-billed curlew detections (white centre) into groups for three delineation approaches. Bottom: Estimated long-billed curlew population trend and 95% confidence interval between 2007 and 2019 from a generalized additive model with year effects.

delineation approach (Figure 4; Appendix S5, Table S5.1). There were differences between groups detected for departure and arrival timing, home range area and number of winter home ranges. The West group from the clustering and expert approaches departed for and arrived from prebreeding and postbreeding migration at least 5 days prior to the other groups. There were substantial differences in the size of home range or stopover area across groups and seasons; notably, the groups with declining population trends had prebreeding migration stopovers approximately two to three times larger than the groups with positive trends for the expert and clustering approaches, but not for the flyway approach. There were also approximately 0.5–1.5 more mean wintering home ranges per individual for those groups across all approaches.

3.5 | Habitat selection

There were more between-group differences in habitat selection revealed by the expert and clustering approaches than the flyway

approach. (Figure 5; Appendix S5, Table S5.2). The proportion of cropland was most strongly selected for across environmental attributes and approaches, with a strong positive selection strength and differences between groups in every season except the breeding season with the expert and clustering approaches. Cropland was particularly important for the intermountain group during migration stopovers and for the Chihuahuan group during the nonbreeding season. In contrast, there was higher selection strength for the proportion of grassland during the breeding season, with stronger selection for the West/Pacific group, but the overall selection strength was much lower than during other seasons. Selection for wetland areas was weak and only occurred during the nonbreeding season.

4 | DISCUSSION

We used satellite tracking data collected across the long-billed curlew's range to compare three approaches for identifying

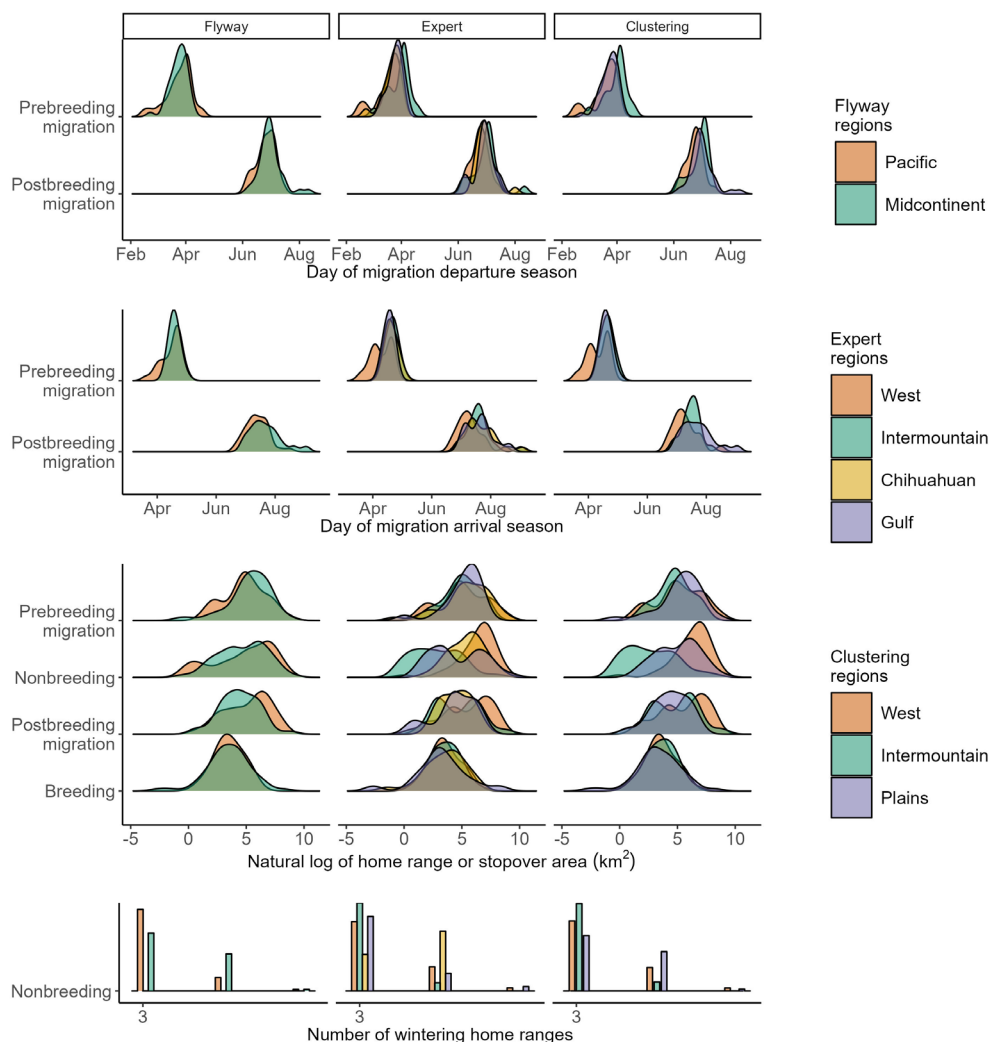


FIGURE 4 Relative distributions of long-billed curlew movement behaviours for three approaches to defining geographic groups across the annual cycle. Density plots or histograms are plotted from values calculated from satellite tracking data of individual birds and y-axes represent counts of individual birds.

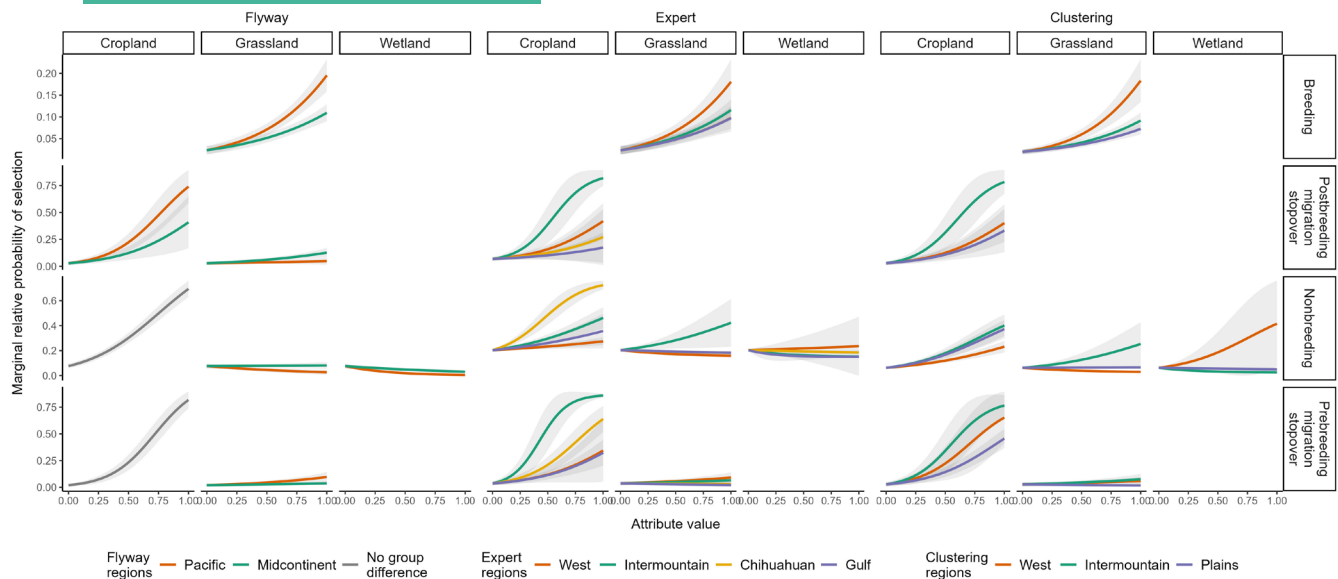


FIGURE 5 Second order habitat selection of long-billed curlews of geographic groups for three approaches to defining groups. Solid lines indicate mean model prediction and shaded areas are 95% confidence intervals of that prediction. 'No group difference' indicates a covariate for which the interaction with group assignment was not included in the best model selected via Akaike's information criterion.

ecologically distinct groups for annual cycle management: flyway grouping, expert grouping and unsupervised clustering based on migratory connectivity. All approaches resulted in groups with different population trends, movement behaviours and second order habitat selection across the annual cycle; however, the flyway delineation approach had weaker migratory connectivity, marginal differences in population trends, and less between-group differences in movement behaviour and habitat selection than the other approaches. The expert and clustering approaches resulted in similar group configuration, estimates of migratory connectivity, movement behaviour and habitat selection; and, the expert approach identified an additional divide between the easternmost grouping, which revealed strongly negative population trends in the Chihuahuan group. We found that western long-billed curlews and those that overwinter in the Chihuahuan desert had significantly negative population trends from 2007 to 2019, while populations that breed and migrate in the intermountain region and those that overwinter along the Gulf of Mexico showed positive trends. Movement behaviours differed most among groups during the migration and the stationary nonbreeding seasons. Habitat selection also differed among groups, particularly for the proportion of cropland at postbreeding and prebreeding migration stopovers.

In our case study, expert-defined groupings revealed a migratory divide that the clustering approach did not identify, which provided additional insight into population declines. As a globally imperilled group (Smith, Binley, et al., 2023; Smith, Smith, et al., 2023), the *Numenius* have received extensive research attention and there is abundant expert opinion available (e.g. Pearce-Higgins et al., 2017); however, expert opinion may not be as readily available or well-informed for other species of conservation concern. Furthermore, the expert groupings we defined were based on distinct geography of the stationary nonbreeding grounds;

many other migratory species occupy a much more homogeneous nonbreeding range and may be more difficult to separate based on expert opinion alone. In these cases of limited expertise or indistinct geographies, unsupervised clustering based on migratory connectivity may be a valuable tool. The unsupervised clustering approach could also be extended to include ecological attributes in addition to geographic coordinates in the K-means clustering, depending on the ecology of the species. Where expert opinion groupings are available, we recommend comparing them to unsupervised clustering, as expert opinion does not always provide accurate results in ecological applications (e.g. Bennett et al., 2023; Seoane et al., 2005; Travers et al., 2023).

Whereas other approaches to group delineation have merits (e.g. genetic clustering, (Ruegg et al., 2017), abundance clustering (Rushing et al., 2015; Schuster et al., 2019), grouping by existing management units (Meehan et al., 2022)), they may not maximize the strength of migratory connectivity. In comparison to flyway grouping, we showed that a clustering approach with three groups maximized estimates and minimized error of migratory connectivity across the annual cycle, despite the presence of only two major flyways in our study region (Pacific and Midcontinent). Identification of this 'hidden' population divide may allow wildlife managers to undertake more targeted conservation actions for different groups with different population trends. For example, conservation of key foraging areas in important prebreeding migration stopover areas (Fellows & Jones, 2009; Sessler, 2013) could be prioritized for the Chihuahuan group if future research reveals that the steep decline of that population is linked to prebreeding migration foraging habitat.

Clustering the data into a range of groupings and then selecting the level that maximized the strength of migratory connectivity also provided a less subjective and biased estimation of migratory connectivity. We could have underestimated breeding to nonbreeding

migratory connectivity by almost 40% if we had used an a priori defined number of groups (0.72 for 3 populations vs. 0.45 for 5 populations) and 15% if we had used flyway groupings (0.62; E. Knight, unpublished data). Other studies have also shown that the strength of migratory connectivity is scale-dependent, both spatially and temporally (Sharp et al., 2022; Vickers et al., 2021). As mentioned, a priori approaches to identifying groups may result in an misestimation of the strength of migratory connectivity (Vickers et al., 2021) and could result in targeting management actions in the wrong place or time.

A final advantage of our group delineation approach is that it allows for quantitative evaluation of differences in population trend across population divides. Migratory connectivity is often associated with differential population trends among populations (Cresswell, 2014; Kramer et al., 2018); however, this link is often anecdotal (e.g. Anderwald, 2021; Morricks, 2022). Strata-specific population trend estimates often do not align with finer-scale estimates, likely because population trend is affected by processes that operate at multiple scales or because the strata are not biologically relevant (Smith, Binley, et al., 2023; Smith, Smith, et al., 2023). Strata-specific trend estimates are therefore sensitive to how monitoring data are clustered. For this reason, we spatially defined population divides for the long-billed curlew and segregated monitoring data by those divides prior to concluding that groups with strong migratory connectivity had differential population trends. The results of our trend analysis, particularly for the expert groupings, were similar to trend estimates from eBird (Fink, Auer, et al., 2023; Fink, Johnston, et al., 2023; Sullivan et al., 2009), with some fine-scale variation that likely reflects small-scale processes instead of more regional processes associated with migratory connectivity.

Collectively, our analysis of group differences in long-billed curlew movement behaviour and habitat selection provides direction for next steps in understanding the differential population trends of those groups. We found differences among groups in the size of stopover area during prebreeding migration, which can be indicative of limitations such as poor foraging habitat that affects survivorship (Calderón et al., 2019) or reproductive success via carry over effects of individuals arriving in poor condition on the breeding grounds (Visser et al., 2012). Carry-over effects could also originate on the nonbreeding grounds, where individuals that spend the nonbreeding season in lower quality habitat are in poorer physiological condition, migrate later and therefore have lower reproductive success (Marra et al., 1998; Paxton & Moore, 2015). Larger nonbreeding home range sizes for the groups with declining populations could also be indicative of such lower quality nonbreeding habitat. Future research could link the groups delineated here (Appendix S4; Knight et al., 2025) to demographic data from the breeding grounds to understand how individual variation in prebreeding migration and nonbreeding behaviour is linked to variation in survivorship and reproductive success.

Group differences in habitat selection strength were particularly notable for the proportion of cropland within the home range,

suggesting there may be differences in resources provided by cropland between groups. During migration in non-coastal areas, long-billed curlews use a variety of sparsely vegetated or shallow, wet habitats for foraging and roosting (Fellows & Jones, 2009; Sesser, 2013). Migration, particularly prebreeding in March and April, can be an unpredictable and energetically intensive time for long-billed curlews and cropland may provide reliable foraging resources (Fellows & Jones, 2009). Indeed, we found probabilities of selection strength for croplands near one for at least one of the groups during both migration seasons. Given that we used just a single cropland category in our analysis, it may be that group differences in selection strength for cropland are driven by differences in crop or irrigation type between regions, with stronger selection for crop types that provide more reliable foraging resources during migration. Alternatively, between-group differences in selection strength could be an indication of a functional response to the proportion of agriculture within each region. Regardless, next steps for long-billed curlew conservation could include partitioning variance in monitoring data of the groups delineated here (Appendix S4; Knight et al., 2025) by environmental attributes, especially cropland foraging habitats (see Sesser, 2013), to statistically evaluate potential population threats (Hallworth et al., 2021; Rushing et al., 2016). This type of analysis may also allow for (1) differences in drivers of population trend across groups to be identified and (2) a longer time period of population dynamics.

If population limitations do occur during migration, protection of long-billed curlew habitat during migration may be complicated by the fact that migratory movement is highly variable and presumably opportunistic. We found stopover use for some individuals was so variable that it was difficult to separate stationary seasons from migratory seasons, with some individuals undertaking their first 'stopover' 100km from their nonbreeding grounds and other individuals flying non-stop to their breeding grounds. We suggest this behaviour may be an adaptation for migration through a region with variable and ephemeral wetland availability, including foraging areas produced via agricultural irrigation (Skagen & Knopf, 1993). This energy-minimizing strategy renders long-billed curlew populations less vulnerable to the loss of a single important stopover site like other shorebird species; however, it also requires coordinated, region-wide conservation planning and policy to prevent habitat loss and degradation along the migration corridor instead of the protection of a limited series of key stopover sites.

We conclude that migratory connectivity is a powerful tool for identifying migratory divides and for creating effective groups that may be useful in wildlife management contexts and should be used in concert with expert opinion when possible. We suggest the clustering approach works particularly well when migratory connectivity is strong, as we found for the long-billed curlew (breeding to nonbreeding MC=0.68). Other shorebird species have been found to also demonstrate strong migratory connectivity, including the willet (MC=0.90; Huysman et al., 2022) and black-bellied plover (Mantel correlation=0.50; A-L. Harrison, unpublished data), and require different conservation strategies across groups (Huysman et al., 2022).

Strong migratory connectivity links individual populations in space and time, which means breeding populations may be vulnerable to the same environmental conditions during other seasons of the annual cycle, but it also makes disentangling differential population trends more straightforward (Cresswell, 2014). For any species, identifying ecologically distinct groups is a vital first step for wildlife management so that population declines can be identified, understood and reversed. Those groupings, including those delineated here (Appendix S4; Knight et al., 2025) can then be used for a variety of applications related to the identification and reversal of population declines. For example, US and Canadian Migratory Bird Habitat Joint Ventures (Giocomo et al., 2012) set habitat conservation objectives based on population groupings for specific priority species (of which the long-billed curlew is included; Vermillion & Lancaster, 2022). We argue that using patterns of migratory connectivity to delineate groups for such management is a particularly effective approach because it allows for quantitative evaluation of the link between migratory connectivity and population trend, can be linked to habitat and behaviour and incorporates the annual cycle into population management, which is increasingly recognized as a key component for the conservation of migratory populations (Marra et al., 2015).

AUTHOR CONTRIBUTIONS

Autumn-Lynn Harrison, Jay Carlisle and Peter P. Marra initiated the manuscript and provided strategic direction. Elly C. Knight led conceptual design, data collation, data analysis and writing. Autumn-Lynn Harrison, Jay Carlisle, Andy J. Boyce, Stephanie Coates and T. Lee Tibbitts provided in-depth feedback on early versions of the manuscript. Jay Carlisle, Andy J. Boyce, David Bradley, Paula Cimprich, Stephanie Coates, Stephen J. Dinsmore, Cory J. Gregory, Joel G. Jorgensen, Jeffrey F. Kelly, David Newstead, Alina Olalla, Larkin A. Powell, Amy L. Scarpignato, T. Lee Tibbitts, Nils Warnock, Walter Wehtje and Autumn-Lynn Harrison led field deployments, managed data collection, contributed to initial project priorities and contributed feedback on the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

AFFILIATIONS

¹Migratory Bird Center, Smithsonian's National Zoo and Conservation Biology Institute, Washington, District of Columbia, USA; ²Department of Biological Sciences, University of Alberta, Edmonton, Alberta, Canada; ³Intermountain Bird Observatory, Boise State University, Boise, Idaho, USA; ⁴Birds Canada, Port Rowan, Ontario, Canada; ⁵School of Biological Sciences, University of Oklahoma, Norman, Oklahoma, USA; ⁶Point Blue Conservation Science, Petaluma, California, USA; ⁷Department of Natural Resource Ecology and Management, Iowa State University, Ames, Iowa, USA; ⁸Field Guides Inc, Austin, Texas, USA; ⁹Nongame Bird Program, Nebraska Game and Parks Commission, Lincoln, Nebraska, USA; ¹⁰Coastal Bend Bays & Estuaries Program, Corpus Christi, Texas, USA; ¹¹Universidad Autónoma de Nuevo León, San Nicolás de los Garza, Mexico; ¹²School of Natural Resources, University of Nebraska-Lincoln, Lincoln, Nebraska, USA; ¹³U.S. Geological Survey, Alaska Science Center, Anchorage, Alaska, USA; ¹⁴Audubon Canyon Ranch, Stinson Beach, California, USA; ¹⁵Ricketts Conservation Foundation, Bondurant, Wyoming, USA and ¹⁶Department of Biology, McCourt School of Public Policy, The Earth Commons Institute, Georgetown University, Washington, District of Columbia, USA

ACKNOWLEDGEMENTS

Our sincere thanks to the many volunteers who assisted with deployment of tags on long-billed curlews across North America, particularly Bob Gill and Gary Page who led the first satellite tracking project on long-billed curlew. Thanks to Adam Smith for insight on strata-specific trend estimation and to anonymous reviewers for excellent feedback that improved our manuscript. Funding for the Migratory Connectivity Project in support of field deployments in Texas and Georgia, United States of America, and Saskatchewan, Canada, as well as a Smithsonian Post-doctoral Fellowship for ECK, was provided by a gift to the Smithsonian from ConocoPhillips Charitable Investments. Funding for field deployments in British Columbia, Canada provided by Environment and Climate Change Canada, the BC Hydro Fish and Wildlife Compensation Program, the Habitat Conservation Trust Foundation, and the Columbia Basin Trust. Funders for IBO: Bureau of Land Management, US Fish and Wildlife Service, Wyoming Game and Fish Department, California Department of Fish and Wildlife, Meg and Bert Raynes Wildlife Fund, Wyoming Governor's office Big Game Licence Coalition, McDanel Land Foundation, Page Family Foundation, The Nature Conservancy and MPG Ranch. Funding for deployments in eastern Montana by Smithsonian were made possible by the generous support of John and Adrienne Mars and American Prairie. Any use of trade, firm or product names is for descriptive purposes only and does not imply endorsement by the US Government.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

Processed movement data and code for analyses available via the Zenodo Repository <https://doi.org/10.5281/zenodo.14607317> (Knight et al., 2025). Raw data for 148 tracked individuals (six of nine projects) included are also available on Movebank (Appendix S1).

ORCID

Elly C. Knight  <https://orcid.org/0000-0002-8578-892X>

David Bradley  <https://orcid.org/0000-0002-5489-6258>

Paula Cimprich  <https://orcid.org/0000-0001-5728-8832>

REFERENCES

- Akaike, H. (1987). Factor analysis and AIC. *Psychometrika*, 52(3), 317–332. <https://doi.org/10.1007/bf02294359>
- Alerstam, T., & Hedenström, A. (1998). The development of bird migration theory. *Journal of Avian Biology*, 29(4), 343. <https://doi.org/10.2307/3677155>
- Ambrosini, R., Møller, A. P., & Saino, N. (2009). A quantitative measure of migratory connectivity. *Journal of Theoretical Biology*, 257(2), 203–211. <https://doi.org/10.1016/j.jtbi.2008.11.019>
- Anderwald, D., Czajka, Ł., Rubacha, S., Zygmunt, M., & Mirski, P. (2021). Autumn migration of ospreys from two distinct populations in Poland reveals partial migratory divide. *Avian Research*, 12(1), 46. <https://doi.org/10.1186/s40657-021-00281-6>

- Aybar, C. (2024). *rgee: R bindings for calling the 'Earth Engine' API*. R package version 1.1.7, <https://r-spatial.github.io/rgee/>, <https://github.com/google/earthengine-api/>, <https://github.com/r-spatial/rgee/>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. C. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bennett, A. F., Haslen, A., White, M., Hollings, T., & Thomson, J. R. (2023). How expert are 'experts'? Comparing expert predictions and empirical data on the use of farmland restoration sites by birds. *Biological Conservation*, 282, 110018. <https://doi.org/10.1016/j.biocon.2023.110018>
- Boere, G. C., & Stroud, D. A. (2006). The flyway concept: What it is and what it isn't. In G. C. Boere, C. A. Galbraith, & D. A. Stroud (Eds.), *Waterbirds around the world* (pp. 40–47). Scottish Natural Heritage.
- Calderón, C. L., Arenas, J. B., Hobson, K. A., & Møller, A. P. (2019). Age-dependent carry-over effects in a long-distance migratory bird. *Scientific Reports*, 9(1), 12032. <https://doi.org/10.1038/s41598-019-47374-3>
- Calenge, C. (2006). The package "adehabitat" for the R software: A tool for the analysis of space and habitat use by animals. *Ecological Modelling*, 197(3–4), 516–519. <https://doi.org/10.1016/j.ecolmodel.2006.03.017>
- CLS America. (2016). *Argos user's manual*. CLS America.
- Coates, S. E. (2018). *Building the full annual cycle picture for long-billed Curlews: Correlates of nest success in the breeding grounds and spatial distribution and site fidelity in the wintering grounds*. Boise State University.
- Cochran, J. F., & Anderson, S. H. (1987). Comparison of habitat attributes at sites of stable and declining long-billed curlew populations. *The Great Basin Naturalist*, 47(3), 459–466.
- Cohen, E. B., Hostetler, J. A., Hallworth, M. T., Rushing, C. S., Sillett, T. S., & Marra, P. P. (2017). Quantifying the strength of migratory connectivity. *Methods in Ecology and Evolution*, 85, 749–812. <https://doi.org/10.1111/2041-210x.12916>
- Cresswell, W. (2014). Migratory connectivity of Palaearctic–African migratory birds and their responses to environmental change: The serial residency hypothesis. *Ibis*, 156(3), 493–510. <https://doi.org/10.1111/ibi.12168>
- Dugger, K., & Dugger, B. (2020). Long-billed curlew (*Numenius americanus*), version 1.0. In A. F. Poole & F. Gill (Eds.), *The birds of the world*. Cornell Lab of Ornithology. <https://doi.org/10.2173/bow.lobcur.01>
- Fellows, S. D., & Jones, S. L. (2009). *Status assessment and conservation action plan for the long-billed curlew (Numenius americanus)*. US Fish and Wildlife.
- Finch, T., Butler, S. J., Franco, A. M. A., & Cresswell, W. (2017). Low migratory connectivity is common in long-distance migrant birds. *Journal of Animal Ecology*, 86(3), 662–673. <https://doi.org/10.1111/1365-2656.12635>
- Finch, T., Saunders, P., Avilés, J. M., Bermejo, A., Catry, I., de la Puente, J., Emmenegger, T., Mardegá, I., Mayet, P., Parejo, D., Račinskis, E., Rodríguez-Ruiz, J., Sackl, P., Schwartz, T., Tiefenbach, M., Valera, F., Hewson, C., Franco, A., & Butler, S. J. (2015). A pan-European, multipopulation assessment of migratory connectivity in a near-threatened migrant bird. *Diversity and Distributions*, 21(9), 1051–1062. <https://doi.org/10.1111/ddi.12345>
- Fink, D., Auer, T., Johnston, A., Strimas-Mackey, M., Ligocki, S., Robinson, O., Hochachka, W., Jaromczyk, L., Crowley, C., Dunham, K., Stillman, A., Davies, I., Rodewald, A., Ruiz-Gutierrez, V., & Wood, C. (2023). *eBird status and trends, data version: 2022; released: 2023*. Cornell Lab of Ornithology. <https://doi.org/10.2173/ebirdst.2022>
- Fink, D., Johnston, A., Strimas-Mackey, M., Auer, T., Hochachka, W. M., Ligocki, S., Jaromczyk, L. O., Robinson, O., Wood, C., Kelling, S., & Rodewald, A. D. (2023). A double machine learning trend model for citizen science data. *Methods in Ecology and Evolution*, 14(9), 2435–2448. <https://doi.org/10.1111/2041-210x.14186>
- Friedl, M., & Sulla-Menasse, D. (2022). *MODIS/Terra+qua land cover type yearly L3 global 500 m SIN grid V061*. NASA EOSDIS Land Processes Distributed Active Archive Center.
- Giocomo, J. J., Gustafson, M., Duberstein, J. N., & Boyd, C. (2012). The role of joint ventures in bridging the gap between research and management. In J. P. Sands, S. J. DeMaso, M. J. Schnupp, & L. A. Brennan (Eds.), *Wildlife science: Connecting research with management* (pp. 239–251). CRC Press.
- Giorgino, T. (2009). Computing and visualizing dynamic time warping alignments in R: The dtw package. *Journal of Statistical Software*, 31(7), 1–24. <https://doi.org/10.18637/jss.v031.i07>
- Hallworth, M. T., Bayne, E., McKinnon, E., Love, O., Tremblay, J. A., Drolet, B., Ibarzabal, J., Wilgenburg, S. V., & Marra, P. P. (2021). Habitat loss on the breeding grounds is a major contributor to population declines in a long-distance migratory songbird. *Proceedings of the Royal Society B*, 288(1949), 20203164. <https://doi.org/10.1098/rspb.2020.3164>
- Hewson, C. M., Thorup, K., Pearce-Higgins, J. W., & Atkinson, P. W. (2016). Population decline is linked to migration route in the common cuckoo. *Nature Communications*, 7(1), 12296. <https://doi.org/10.1038/ncomms12296>
- Hobson, K. A., Kardynal, K. J., Wilgenburg, S. L. V., Albrecht, G., Salvadori, A., Cadman, M. D., Liechti, F., & Fox, J. W. (2015). A continent-wide migratory divide in north American breeding barn swallows (*Hirundo rustica*). *PLoS One*, 10(6), e0129340. <https://doi.org/10.1371/journal.pone.0129340>
- Huysman, A. E., Cooper, N. W., Smith, J. A., Haig, S. M., Heath, S. A., Johnson, L., Olson, E., Regan, K., Wilson, J. K., & Marra, P. P. (2022). Strong migratory connectivity indicates willets need subspecies-specific conservation strategies. *Ornithological Applications*, 124, duac015. <https://doi.org/10.1093/ornithapp/duac015>
- Iwamura, T., Possingham, H. P., Chadès, I., Minton, C., Murray, N. J., Rogers, D. I., Treml, E. A., & Fuller, R. A. (2013). Migratory connectivity magnifies the consequences of habitat loss from sea-level rise for shorebird populations. *Proceedings of the Royal Society B: Biological Sciences*, 280(1761), 20130325. <https://doi.org/10.1098/rspb.2013.0325>
- Johnson, D. H. (1980). The comparison of usage and availability measurements for evaluating resource preference. *Ecology*, 61(1), 65–71.
- Jonsen, I. D., Patterson, T. A., Costa, D. P., Doherty, P. D., Godley, B. J., Grecian, W. J., Guinet, C., Hoerner, X., Kienle, S. S., Robinson, P. W., Votier, S. C., Whiting, S., Witt, M. J., Hindell, M. A., Harcourt, R. G., & McMahon, C. R. (2020). A continuous-time state-space model for rapid quality control of argos locations from animal-borne tags. *Movement Ecology*, 8(1), 31. <https://doi.org/10.1186/s40462-020-00217-7>
- Justen, H., Lee-Yaw, J. A., & Delmore, K. E. (2021). Reduced habitat suitability and landscape connectivity in a songbird migratory divide. *Global Ecology and Biogeography*, 30(10), 2043–2056. <https://doi.org/10.1111/geb.13367>
- Kays, R., Davidson, S. C., Berger, M., Bohrer, G., Fiedler, W., Flack, A., Hirt, J., Hahn, C., Gauggel, D., Russell, B., Kölzsch, A., Lohr, A., Partecke, J., Quetting, M., Safi, K., Scharf, A., Schneider, G., Lang, I., Schaeuffelhut, F., ... Wikelski, M. (2022). The Movebank system for studying global animal movement and demography. *Methods in Ecology and Evolution*, 13(2), 419–431. <https://doi.org/10.1111/2041-210x.13767>
- Knight, E. C., Carlisle, J., Boyce, A. J., Bradley, D., Cimprich, P., Coates, S., Dinsmore, S. J., Gregory, C. J., Jorgensen, J. G., Kelly, J. F., Newstead, D., Olalla, A., Powell, L. A., Scarpignato, A. L., Tibbitts, T. L., Warnock, N., Wehtje, W., Marra, P. P., & Harrison, A.-L. (2025). Data and code from: Delineating ecologically distinct groups for annual cycle management of a declining shorebird. *Zenodo Repository*, <https://doi.org/10.5281/zenodo.14607317>
- Knight, S. M., Bradley, D. W., Clark, R. G., Gow, E. A., Bélisle, M., Berzins, L. L., Blake, T., Bridge, E. S., Burke, L., Dawson, R. D., Dunn, P. O.,

- Garant, D., Holroyd, G. L., Hunsell, D. J. T., Lansdorp, O., Laughlin, A. J., Leonard, M. L., Pelletier, F., Shutler, D., ... Norris, D. R. (2018). Constructing and evaluating a continent-wide migratory songbird network across the annual cycle. *Ecological Monographs*, 88(3), 445–460. <https://doi.org/10.1002/ecm.1298>
- Kramer, G. R., Andersen, D. E., Buehler, D. A., Wood, P. B., Peterson, S. M., Lehman, J. A., Aldinger, K. R., Bulluck, L. P., Harding, S., Jones, J. A., Loegering, J. P., Smalling, C., Vallender, R., & Streby, H. M. (2018). Population trends in Vermivora warblers are linked to strong migratory connectivity. *Proceedings of the National Academy of Sciences of the United States of America*, 115(14), E3192–E3200. <https://doi.org/10.1073/pnas.1718985115>
- Lott, C. A., & Smith, J. P. (2006). A geographic-information-system approach to estimating the origin of migratory raptors in North America using stable hydrogen isotope ratios in feathers. *The Auk*, 123(3), 822–835.
- Manly, B., McDonald, L. L., Thomas, D. L., McDonald, T. L., & Erickson, W. P. (2002). *Resource selection by animals* (2nd ed.). Kluwer Academic Publishers.
- Marra, P. P., Cohen, E., Harrison, A.-L., Studds, C. E., & Webster, M. (2018). *Chapter 20761—Migratory connectivity*. Elsevier Ltd. <https://doi.org/10.1016/b978-0-12-809633-8.20761-9>
- Marra, P. P., Cohen, E. B., & Loss, S. R. (2015). A call for annual cycle research in animal ecology. *Biology Letters*, 11(8), 20150552. <https://doi.org/10.1098/rsbl.2015.0552>
- Marra, P. P., Hobson, K. A., & Holmes, R. T. (1998). Linking winter and summer events in a migratory bird by using stable-carbon isotopes. *Science*, 282(5395), 1884–1886. <https://doi.org/10.1126/science.282.5395.1884>
- Meehan, T. D., Saunders, S. P., DeLuca, W. V., Michel, N. L., Grand, J., Deppe, J. L., Jimenez, M. F., Knight, E. J., Seavy, N. E., Smith, M. A., Taylor, L., Witko, C., Akresh, M. E., Barber, D. R., Bayne, E. M., Beasley, J. C., Belant, J. L., Bierregaard, R. O., Bildstein, K. L., ... Wilsey, C. B. (2022). Integrating data types to estimate spatial patterns of avian migration across the Western Hemisphere. *Ecological Applications*, 32, e2679. <https://doi.org/10.1002/eap.2679>
- Michelot, T., Langrock, R., & Patterson, T. A. (2016). moveHMM: An R package for the statistical modelling of animal movement data using hidden Markov models. *Methods in Ecology and Evolution*, 7(11), 1308–1315. <https://doi.org/10.1111/2041-210X.12578>
- Møller, A. P., & Szép, T. (2011). The role of parasites in ecology and evolution of migration and migratory connectivity. *Journal of Ornithology*, 152(Suppl 1), 141–150. <https://doi.org/10.1007/s10336-010-0621-x>
- Morrison, Z. N., Lilleyman, A., Fuller, R. A., Bush, R., Coleman, J. T., Garnett, S. T., Gerasimov, Y. N., Jessop, R., Ma, Z., Maglio, G., Minton, C. D. T., Syroechkovskiy, E., & Woodworth, B. K. (2022). Differential population trends align with migratory connectivity in an endangered shorebird. *Conservation Science and Practice*, 4(1), e594. <https://doi.org/10.1111/csp2.594>
- Naimi, B., Hamm, N. A. S., Groen, T. A., Skidmore, A. K., & Toxopeus, A. G. (2013). Where is positional uncertainty a problem for species distribution modelling? *Ecography*, 37(2), 191–203. <https://doi.org/10.1111/j.1600-0587.2013.00205.x>
- Olalla-Kerstupp, A., Ruiz-Aymá, G., González-Rojas, J. I., & Guzmán-Velasco, A. (2015a). Using satellite telemetry to identify long-billed curlew winter habitat use in the southeastern corner of the Chihuahuan Desert (Mexico). *The American Midland Naturalist*, 174(1), 117–131. <https://doi.org/10.1674/0003-0031-174.1.117>
- Olalla-Kerstupp, A., Ruiz-Aymá, G., González-Rojas, J. I., & Guzmán-Velasco, A. (2015b). High fidelity to wintering, stop-over and breeding sites shown by a long-billed curlew *Numenius americanus* tracked with satellite telemetry on migratory flights across North America. *Bird Study*, 62(4), 556–560. <https://doi.org/10.1080/00063657.2015.1073678>
- Page, G. W., Warnock, N., Tibbitts, T. L., Jorgensen, D., Hartman, C. A., & Stenzel, L. E. (2014). Annual migratory patterns of long-billed curlews in the American West. *The Condor*, 116(1), 50–61. <https://doi.org/10.1650/condor-12-185-r2.1>
- Paxton, K. L., & Moore, F. R. (2015). Carry-over effects of winter habitat quality on en route timing and condition of a migratory passerine during spring migration. *Journal of Avian Biology*, 46(5), 495–506. <https://doi.org/10.1111/jav.00614>
- Pearce-Higgins, J. W., Brown, D. J., Douglas, D. J. T., Alves, J. A., Bellio, M., Bocher, P., Buchanan, G. M., Clay, R. P., Conklin, J., Crockford, N., Dann, P., Elts, J., Friss, C., Fuller, R. A., Gill, J. A., Gosbell, K., Johnson, J. A., Maequez-Ferrando, R., Masero, J. A., ... Verkuil, Y. I. (2017). A global threats overview for Numeniini populations: Synthesising expert knowledge for a group of declining migratory birds. *Bird Conservation International*, 27(1), 6–34. <https://doi.org/10.1017/s0959270916000678>
- Pfaffel, O. (2020). *Clustlmp: An R package for k-means clustering with build-in missing data imputation*. <https://doi.org/10.13140/RG.2.2.20143.36007>
- Procházka, P., Brlík, V., Yohannes, E., Meister, B., Auerswald, J., Ilieva, M., & Hahn, S. (2018). Across a migratory divide: Divergent migration directions and non-breeding grounds of Eurasian reed warblers revealed by geolocators and stable isotopes. *Journal of Avian Biology*, 49(6), e01769. <https://doi.org/10.1111/jav.01769>
- R Core Team. (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Roberts, A., Scarpignato, A. L., Huysman, A., Hostetler, J. A., & Cohen, E. B. (2023). Migratory connectivity of north American waterfowl across administrative flyways. *Ecological Applications*, 33(3), e2788. <https://doi.org/10.1002/eap.2788>
- Ruegg, K. C., Anderson, E. C., Harrigan, R. J., Paxton, K. L., Kelly, J. F., Moore, F., & Smith, T. B. (2017). Genetic assignment with isotopes and habitat suitability (GAIAH), a migratory bird case study. *Methods in Ecology and Evolution*, 8(10), 1241–1252. <https://doi.org/10.1111/2041-210X.12800>
- Ruegg, K. C., & Smith, T. B. (2002). Not as the crow flies: A historical explanation for circuitous migration in Swainson's thrush (*Catharus ustulatus*). *Proceedings of the Royal Society B: Biological Sciences*, 269(1498), 1375–1381. <https://doi.org/10.1098/rspb.2002.2032>
- Rushing, C. S., Ryder, T. B., & Marra, P. P. (2016). Quantifying drivers of population dynamics for a migratory bird throughout the annual cycle. *Proceedings of the Royal Society B: Biological Sciences*, 283(1823), 20152846. <https://doi.org/10.1098/rspb.2015.2846>
- Rushing, C. S., Ryder, T. B., Scarpignato, A. L., Saracco, J. F., & Marra, P. P. (2015). Using demographic attributes from long-term monitoring data to delineate natural population structure. *Journal of Applied Ecology*, 53(2), 491–500. <https://doi.org/10.1111/1365-2664.12579>
- Sauer, J. R., Niven, D. K., Hines, J. E., Ziolkowski, D. J., Jr., Pardieck, K. L., Fallon, J. E., & Link, W. A. (2017). *The North American breeding bird survey, results and analysis 1966–2015. Version 2.07.2017*.
- Schuster, R., Wilson, S., Rodewald, A. D., Arcese, P., Fink, D., Auer, T., & Bennett, J. R. (2019). Optimizing the conservation of migratory species over their annual cycle. *Nature Communications*, 10, 1754. <https://doi.org/10.1038/s41467-019-09723-8>
- Seoane, J., Bustamante, J., & Díaz-Delgado, R. (2005). Effect of expert opinion on the predictive ability of environmental models of bird distribution. *Conservation Biology*, 19, 512–522. <https://doi.org/10.1111/j.1523-1739.2005.00364.x>
- Sesser, K. A. (2013). *Habitat and space use by long-billed curlews (Numenius americanus) in California's central valley*. Humboldt State University.
- Sharp, A., Rushing, C., Contina, A., Ruiz-Gutierrez, V., Sillett, S., Bridge, E., Besozzi, E., Muller, J., Kelly, J., & Given, A. (2022). *Spatial and temporal scale-dependence of the strength of migratory connectivity in a North American passerine*. <https://doi.org/10.21203/rs.3.rs-1483049/v1>

- Skagen, S. K., & Knopf, F. L. (1993). Toward conservation of midcontinental shorebird migrations. *Conservation Biology*, 7(3), 533–541. <https://doi.org/10.1046/j.1523-1739.1993.07030533.x>
- Smith, A. C., Binley, A. D., Daly, L., Edwards, B. P. M., Ethier, D., Frei, B., Iles, D., Meehan, T. D., Michel, N. L., & Smith, P. A. (2023). Spatially explicit Bayesian hierarchical models improve estimates of avian population status and trends. *Ornithological Applications*, 126, duad056. <https://doi.org/10.1093/ornithapp/duad056>
- Smith, A. C., & Edwards, B. P. M. (2020). North American Breeding Bird Survey status and trend estimates to inform a wide range of conservation needs, using a flexible Bayesian hierarchical generalized additive model. *Ornithological Applications*, 123(1), 1–16. <https://doi.org/10.1093/ornithapp/duaa065>
- Smith, P. A., Smith, A. C., Andres, B., Francis, C. M., Harrington, B., Friis, C., Morrison, R. I. G., Paquet, J., Winn, B., & Brown, S. (2023). Accelerating declines of North America's shorebirds signal the need for urgent conservation action. *Ornithological Applications*, 125(2), 1–14. <https://doi.org/10.1093/ornithapp/duad003>
- Strimas-Mackey, M., Ligocki, S., Auer, T., & Fink, D. (2023). ebirdst: Access and analyze eBird status and trends data products. R package version 3.2022.0, <https://ebird.github.io/ebirdst/>
- Sullivan, B. L., Wood, C. L., Iliff, M. J., Bonney, R. E., Fink, D., & Kelling, S. (2009). eBird: A citizen-based bird observation network in the biological sciences. *Biological Conservation*, 142(10), 2282–2292. <https://doi.org/10.1016/j.biocon.2009.05.006>
- Travers, S. K., Dorrough, J., Shannon, I., Val, J., Scott, M. L., Moutou, C. J., & Oliver, I. (2023). The importance of expert selection when identifying threatened ecosystems. *Conservation Biology*, 37(6), e14151. <https://doi.org/10.1111/cobi.14151>
- U.S. Fish and Wildlife Service. (2019). *Adaptive harvest management: 2020 hunting season*. U.S. Department of Interior.
- Vermillion, W. G., & Lancaster, J. D. (2022). *Habitat objectives for priority Gulf Coast joint venture shorebird species*. Gulf Coast Joint Venture. 10 pp + appendices. https://www.researchgate.net/publication/371566660_Habitat_Objectives_for_Priority_Gulf_Coast_Joint_Venture_Shorebird_Species
- Vickers, S. H., Franco, A. M. A., & Gilroy, J. J. (2021). Sensitivity of migratory connectivity estimates to spatial sampling design. *Movement Ecology*, 9(1), 16. <https://doi.org/10.1186/s40462-021-00254-w>
- Visser, M. E., te Marvelde, L., & Lof, M. E. (2012). Adaptive phenological mismatches of birds and their food in a warming world. *Journal of Ornithology*, 153(Suppl 1), 75–84. <https://doi.org/10.1007/s10336-011-0770-6>
- Warnock, N. (2010). Stopping vs. staging: The difference between a hop and a jump. *Journal of Avian Biology*, 41(6), 621–626. <https://doi.org/10.1111/j.1600-048x.2010.05155.x>
- Webster, M. S., Marra, P. P., Haig, S. M., & Bensch, S. (2002). Links between worlds: Unraveling migratory connectivity. *Trends in Ecology & Evolution*, 17(2), 76–83. [https://doi.org/10.1016/s0169-5347\(01\)02380-1](https://doi.org/10.1016/s0169-5347(01)02380-1)
- Ziolkowski, D. J., Lutmerding, M., English, W. B., Aponte, V. I., & Hudson, M.-A. R. (2023). *North American breeding bird survey dataset 1966–2022: U.S. Geological Survey data release*. <https://doi.org/10.5066/P9GS9K64>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1: Metadata for individual long-billed curlew satellite tracking datasets.

Appendix S2: Code used for analyses.

Appendix S3: Detailed movement data processing methods.

Appendix S4: Group polygons per season.

Appendix S5: AIC tables for model selection.

How to cite this article: Knight, E. C., Carlisle, J., Boyce, A. J., Bradley, D., Cimprich, P., Coates, S., Dinsmore, S. J., Gregory, C. J., Jorgensen, J. G., Kelly, J. F., Newstead, D., Olalla, A., Powell, L. A., Scarpignato, A. L., Tibbitts, T. L., Warnock, N., Wehtje, W., Marra, P. P., & Harrison, A.-L. (2025). Delineating ecologically distinct groups for annual cycle management of a declining shorebird. *Journal of Applied Ecology*, 62, 1152–1165. <https://doi.org/10.1111/1365-2664.14885>